

August 4, 2026

Dear Hiring Team,

I am writing to apply for the Machine Learning Engineer position with Kogniz. With a Master of Science in Computer Science from the University of Calgary and applied experience in Python, C++, PyTorch, computer vision, deep learning, and scalable research software, I am excited by the opportunity to contribute to state-of-the-art vision and machine learning products in a focused engineering team.

My graduate research has centered on building machine learning and data-processing systems for large geospatial and Earth observation datasets. In my digital elevation model super-resolution project, I designed Python and PyTorch workflows for data collection, preprocessing, model training, and evaluation. I trained deep ResNet models for 4x terrain reconstruction and compared model performance using RMSE, slope, TRI, TPI, and wavelet-based metrics. This work required careful experimentation, debugging, baseline comparison, and practical judgment about whether models generalized beyond convenient random splits.

I also developed C++ and Python software for scalable processing, storage, and retrieval of satellite imagery through my work with BigGeo, Mitacs, and the University of Calgary. I built DGGS-based encoding pipelines, designed tensor-based storage for multiresolution retrieval, and improved processing performance with C++ multithreading. These projects strengthened the engineering habits I would bring to Kogniz: writing modular code, testing ideas through working prototypes, documenting technical decisions, and making research outputs usable by others.

The Kogniz role is especially interesting to me because it combines computer vision, deep learning frameworks, Python and C++, and close collaboration with expert developers. My background does not come from a traditional product computer-vision team, but it does include hands-on deep learning, scientific imaging, model evaluation, C++/Python engineering, and clear technical communication. I am confident I can contribute quickly while continuing to deepen my production computer-vision experience.

Thank you for considering my application. I would welcome the opportunity to discuss how my machine learning research, software engineering experience, and interest in practical computer vision systems can contribute to Kogniz.

Sincerely,

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